



Swiss Institute of
Bioinformatics

DAY 1, PART 5

Using AlphaFold2 via ColabFold

Diana Rapota, Rok Breznikar, Xavier Robin, Janani Durairaj

23-24 June 2026



ColabFold introduction

nature|methods

BRIEF COMMUNICATION

<https://doi.org/10.1038/s41592-022-01488-1>



OPEN

ColabFold: making protein folding accessible to all

Milot Mirdita ^{1,10} ✉, Konstantin Schütze ², Yoshitaka Moriwaki ^{3,4}, Lim Heo ⁵,
Sergey Ovchinnikov ^{6,7,10} ✉ and Martin Steinegger ^{2,8,9,10} ✉

ColabFold offers accelerated prediction of protein structures and complexes by combining the fast homology search of MMseqs2 with AlphaFold2 or RoseTTAFold. ColabFold's 40–60-fold faster search and optimized model utilization enables prediction of close to 1,000 structures per day on a server with one graphics processing unit. Coupled with Google Colaboratory, ColabFold becomes a free and accessible platform for protein folding. ColabFold is open-source software available at <https://github.com/sokrypton/ColabFold> and its novel environmental databases are available at <https://colabfold.mmseqs.com>.

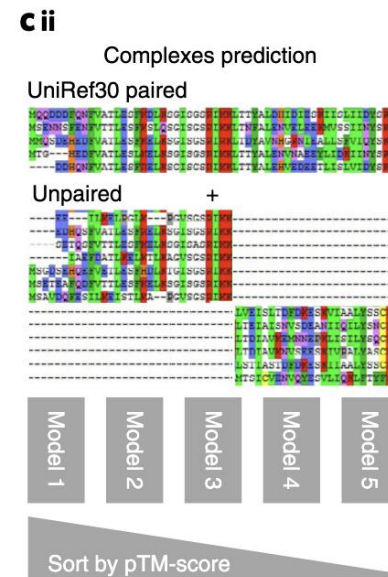
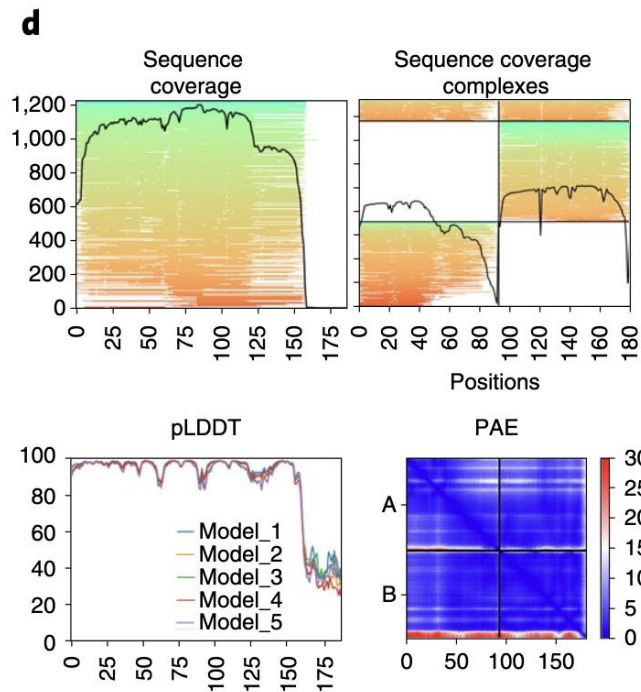
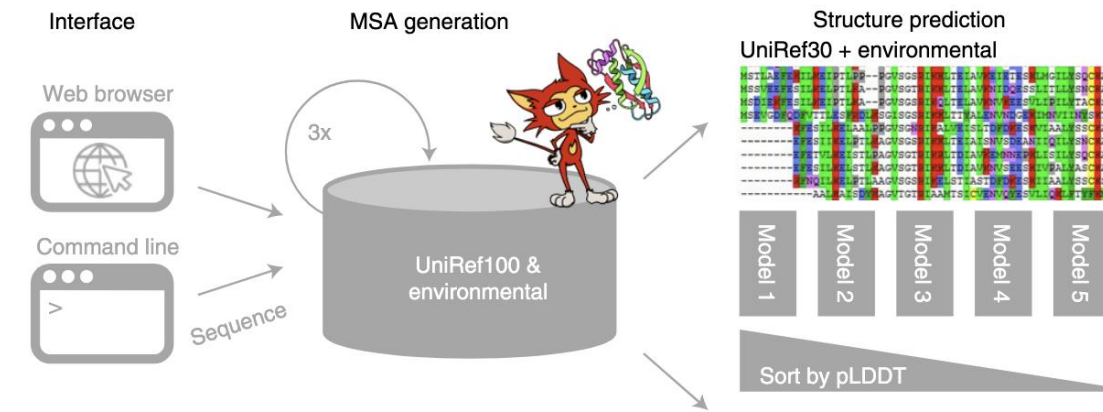
protein sizes of ~1,000 residues. For these, however, the MSA generation dominates the overall run time.

To enable researchers without these resources to use AlphaFold2, independent solutions based on Google Colaboratory were developed. Colaboratory is a proprietary version of Jupyter Notebook hosted by Google. It is accessible for free to logged-in users and includes access to powerful GPUs. Concurrently, Tunyasuvunakool et al.⁹ developed an AlphaFold2 Jupyter Notebook for Google Colaboratory (referred to as AlphaFold-Colab), for which the input MSA is built by searching with HMMer against the UniProt Reference Clusters (UniRef90) and an eightfold-reduced environ-

- ColabFold speeds up monomeric and batch prediction
- Available as Jupyter Notebook for running in Google Colab or locally



ColabFold structure overview



- ColabFold has two interfaces (web and local).
- MSA is generated on the MSA server (there is a limit for multimeric complexes, but you can generate MSA locally with **colabfold_search**).
- Structure prediction in Google Colab or locally.



ColabFold GitHub

ColabFold GitHub is the source of user installation guides/recommendations, updates, info, etc.

For single predictions use: AlphaFold2_mmseqs2

For batch predictions use: AlphaFold2_batch

Useful links:

User guide:

<https://doi.org/10.1038/s41596-024-01060-5>

ColabFold GitHub:

<https://github.com/sokrypton/ColabFold>



[README](#) [Contributing](#) [MIT license](#)

ColabFold - v1.6.1

For details of what was changed in v1.6.1, see [change log!](#)



Making Protein folding accessible to all via Google Colab!

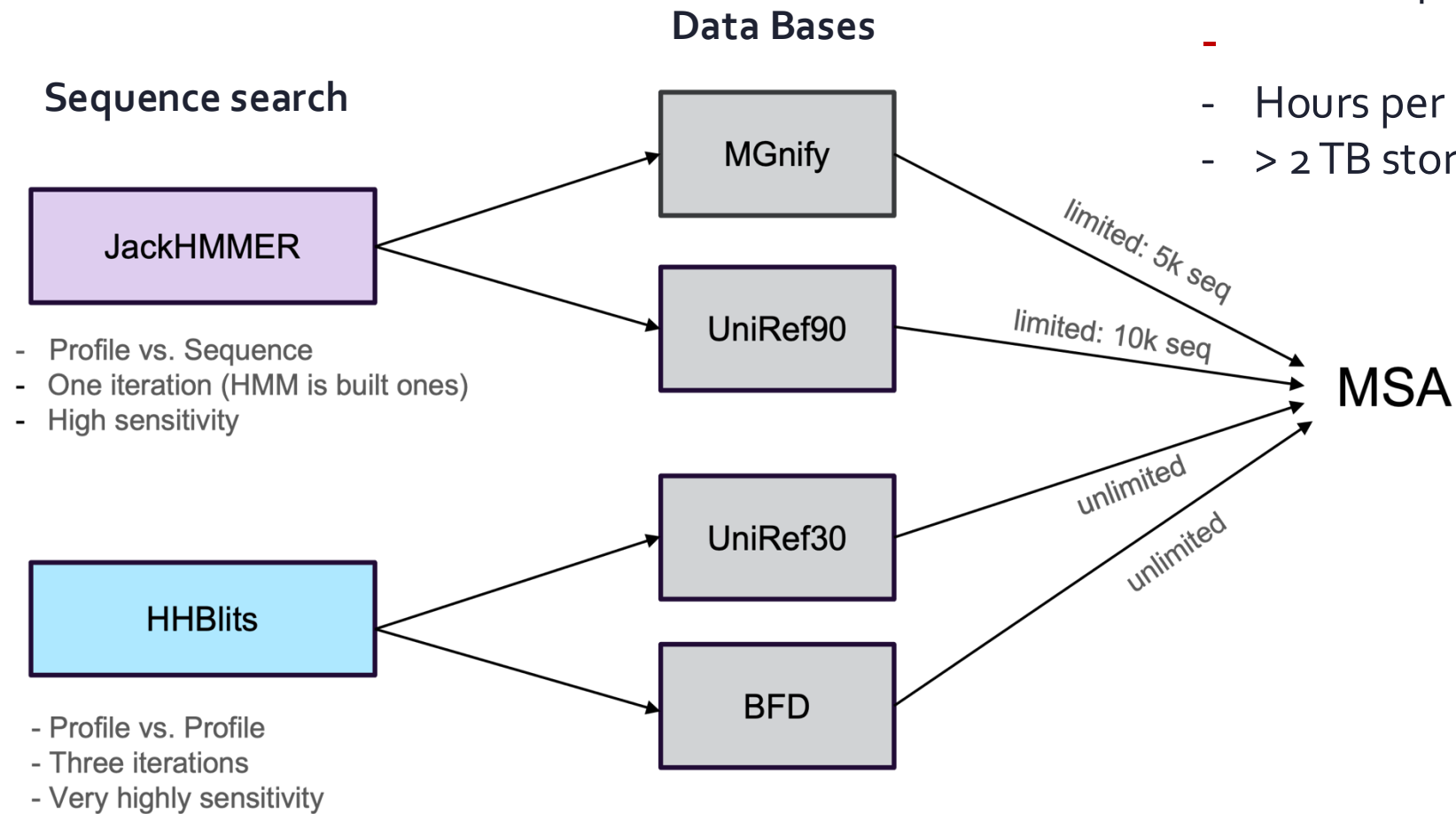
Notebooks	monomers	complexes	mmseqs2	jackhmmer	templates
AlphaFold2_mmseqs2	Yes	Yes	Yes	No	Yes
AlphaFold2_batch	Yes	Yes	Yes	No	Yes
AlphaFold2 (from Deepmind)	Yes	Yes	No	Yes	No
relax_amber (relax input structure)					
ESMFold	Yes	Maybe	No	No	No
BETA (in development) notebooks					
RoseTTAFold2	Yes	Yes	Yes	No	WIP
Boltz	Yes	Yes	Yes	No	No
BioEmu	Yes	No	Yes	No	No
OmegaFold	Yes	Maybe	No	No	No
AlphaFold2_advanced_v2 (new experimental notebook)	Yes	Yes	Yes	No	Yes

Check the wiki page [old retired notebooks](#) for unsupported notebooks.



AlphaFold2 vs ColabFold: MSA construction part

How AF2 builds MSA?



+

- Sensitive homology detection
- Metagenomics databases capture protein families poorly represented in UniRef

-

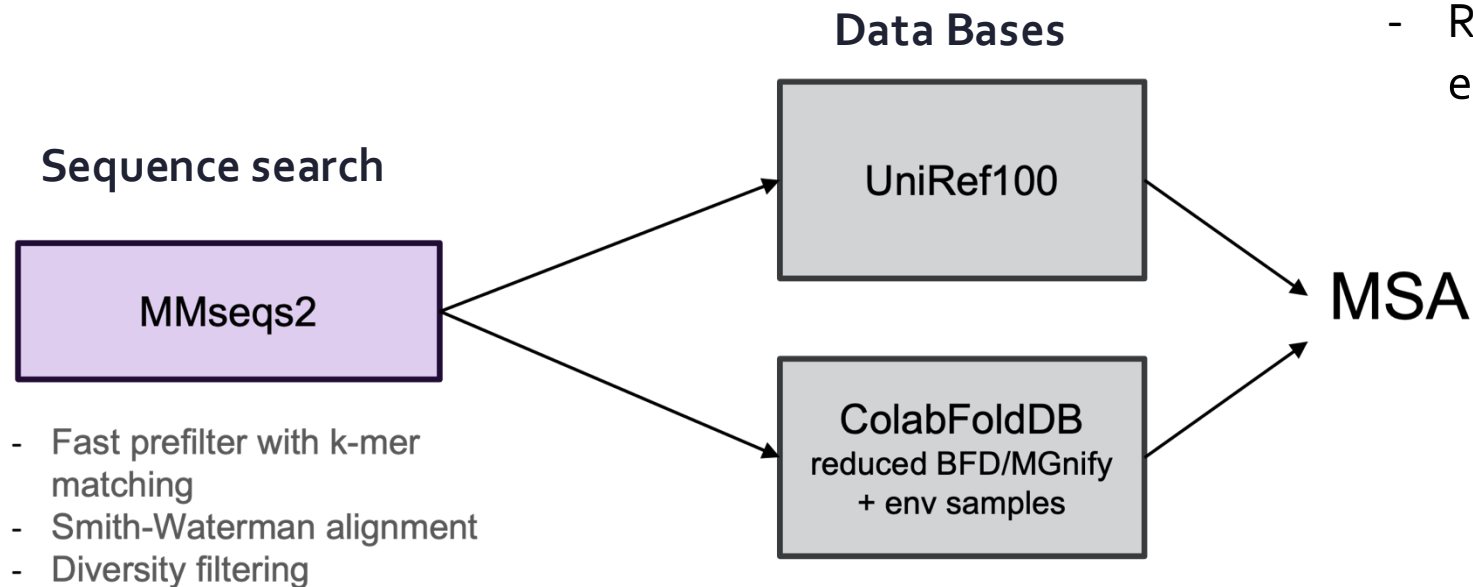
- Hours per protein for database searches
- > 2 TB storage requirement for DBs



AlphaFold2 vs ColabFold: MSA construction

How ColabFold builds MSA?

- MMseqs2 instead of JackHMMER and HHBlits
- Optimized MSA generation by MMseqs2 (diverse but small)
- Redundancy-reduced version of the environmental DBs (ColabFoldDB)





AlphaFold2 vs ColabFold: Structure prediction

- Same architecture
- Gain in speed through model inference:
 - Avoid recompilation of AF2 models
 - Avoid recompiling during batch computation (same input size for all targets with `make_fixed_size`)
 - Customize recycle count
 - Early stop criterion



When it is a good idea to use ColabFold instead of AF2?

- Overall, ColabFold is a good, faster version of AF2 which gives you a lot of flexibility
- The main difference is only about MSA generation and the searched databases
- In the original paper of ColabFold (Mirdita M. et al., 2022), the authors claimed that they gained in speed without losing in sensitivity (but benchmark set was only 65 targets containing 91 domains)

ColabFold is helpful when:

- You don't have the resources to run predictions locally
- You want faster turnaround (the most time-consuming step, MSA construction, is faster)
- For teaching and early exploratory steps
- It is under active development



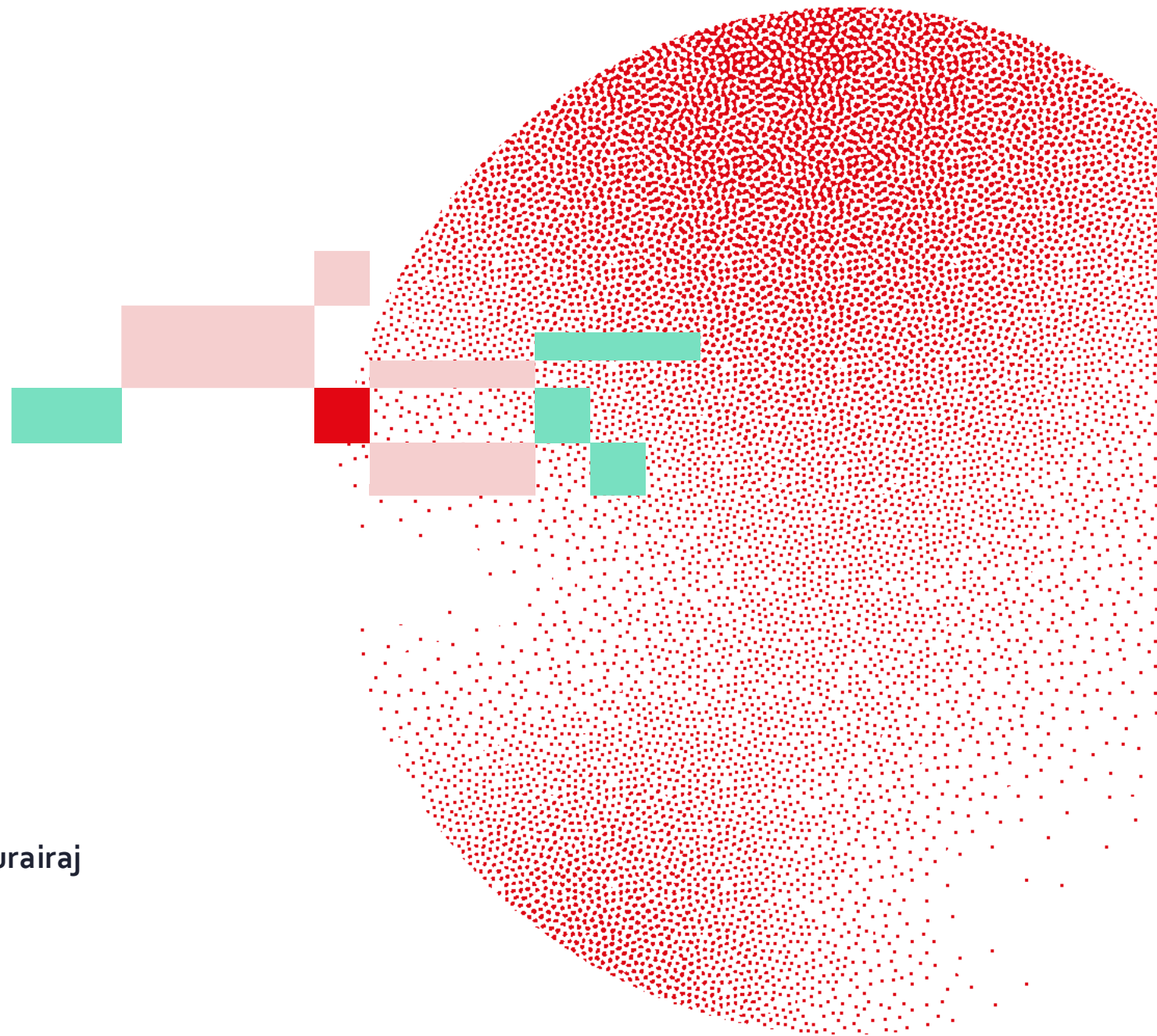
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DAY 1, PART 6

ColabFold tutorial Exercise 1

Diana Rapota, Rok Breznikar, Xavier Robin, Janani Durairaj

23-24 June 2026





Schedule

Day 1

Proteins, AlphaFold2 lecture

09:00 - 10:30

Coffee break

10:30 - 11:00

ColabFold lecture, tutorial 1

11:00 - 12:30

Lunch

12:30 - 13:30

ColabFold tutorial 2, Mol*

13:30 - 15:00

Coffee break

15:00 - 15:30

ColabFold tutorial 3

15:30 - 17:00



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DAY 1, PART 7

Visualisation of structures with Mol*

Diana Rapota, Rok Breznikar, Xavier Robin, Janani Durairaj

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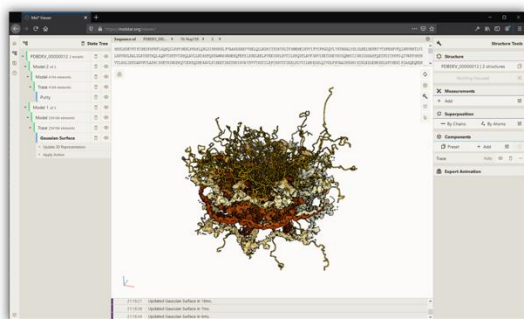
Mol* introduction

Mol* website – molstar.org

Alternatively, use PyMOL or Chimera



Mol* (*/ˈmolstar/*) is a modern web-based open-source toolkit for visualisation and analysis of large-scale molecular data



Open Mol* Viewer

Viewer Documentation

Issues & Feedback

High-performance graphics and data handling of the Mol* Viewer allow users to simultaneously visualise up to hundreds of (superimposed) protein structures, play molecular dynamics trajectories, render cell-level models at atomic detail with tens of millions of atoms, or display huge models obtained by I/HM such as the Nuclear Pore Complex.

Let's get familiar with basic commands in Mol*

Go to Mol* Viewer and upload the suggested structure from the PDB

Mol* tutorial with the exercises are in the **doc**

Source	PDB
PDB Id(s)	1tqn

Options

Apply

The project is an open collaboration started by

Mol* introduction

Reset View

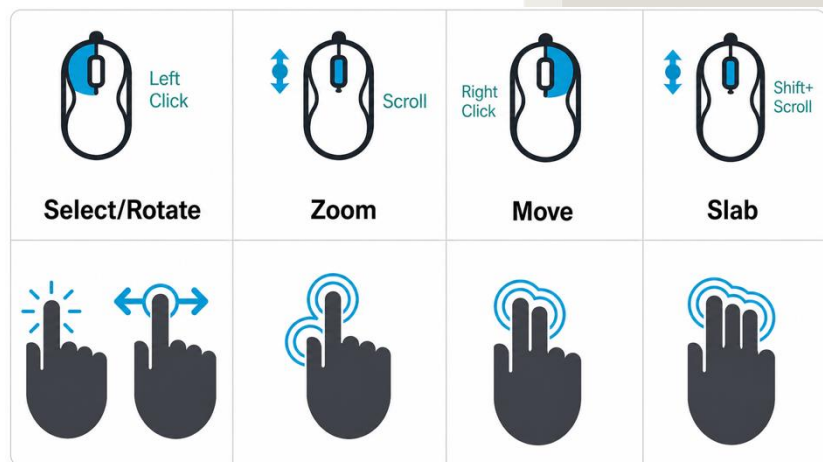
Press *t*

Spin Animation

Press *i*

(to disable press *i* again)

To Navigate the 3D Canvas



Sequence Panel

3D Canvas

Main Menu

Control Panel

Toggle Menu

Log Panel

Structure Tools

Structure

1TQN | Crystal Structure of Huma...

Type Assembly

Asm Id 1: Author Defined As...

Dynamic Bonds × Off

Measurements

+ Add

Quick Styles

Apply Representation

Default Cartoon Spacefill Surface

Apply Style

Default Illustrative

Procedural Animation

Apply Wiggle

Dynamics Uncertainty Clear

Components 1TQN

Preset + Add

Polymer Cartoon

Ligand Ball & Stick

Water Ball & Stick

Unit Cell 1 2 2 2

Volume Streaming 1TQN

✓ Enable

Assembly Symmetry 1TQN

✓ Enable

Sequence of 1TQN | Crystal S Chain 1: cytochrome P A

MALYGTSHGLFKLIGPPTLPFLGNILSYHKGFCMFDECHKYGVWGFYDGOQVLAITDPDMIKTVLVKECYSVFNNRRPFGVGMKSAISIAEDEEWKRLSLLSPTFTSGKLKEMVPIIAQ

YGDVLRNLRRAETGKPVTKDVFAYSDVITSTSGVNIIDSLNPNQDPFVENTKKLRFDLDPFELSITVFFFLIPILEVLNICVFREVTNFLRKSVMKESRLDTQKHRVDLQLMIDSONS

KETESHKALCDLELVAQSIIFIFAGYETTSSVLSFIMVELATHPDVQKLEEDAVLPNKAPPTDYLQMEYLDVNVNLTLLRFLPIAMRLERVCKKDVINGMFI PKGVVVMIPSYALHRDPKYWTEP

15:04:39 Created Ball & Stick in 10ms.

15:04:39 Created Ball & Stick in 2ms.

15:04:39 Updated Structure Focus Representation in 0ms.



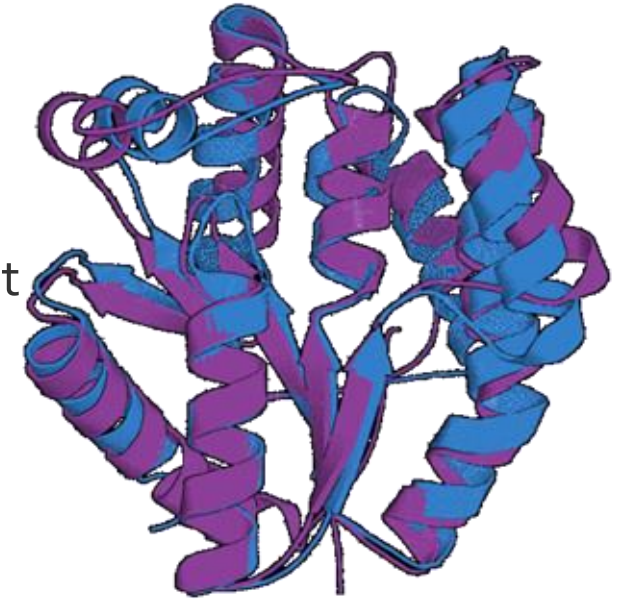
Structure Alignment

What is Structural Alignment?

Structure alignment establishes residue-residue correspondences between macromolecular structures based on optimal superposition of their 3D conformation

When is Structure Alignment useful?

- Structure is more conserved than sequence → alignment helps detect evolutionary relationships where sequence-based methods struggle due to low sequence identity
- Pairwise comparison reveals conformational changes upon ligand binding
- Analysis of structural variation within evolutionary families and identification of common structural domains
- Determining prediction quality



High structural similarity

PDB ids: 4S35, 1NMY

TM-score: 0.82

Sequence identity: 28%



Structure Alignment

In Mol* there two options for structural alignment:

RMSD-based Superposition – rigidly rotates and translates one structure onto another to minimize the RMSD of paired atoms, sequence-guided

RMSD-score: good value ~ 2 Å or less, protein length dependent

TM-align Superposition – sequence-independent alignment algorithm

TM-score: protein length independent, only aligned residues included

Scores < 0.2 usually indicate that the proteins are unrelated while those > 0.5 generally have the same protein fold



Mol* Exercises

Exercise 1 (together)

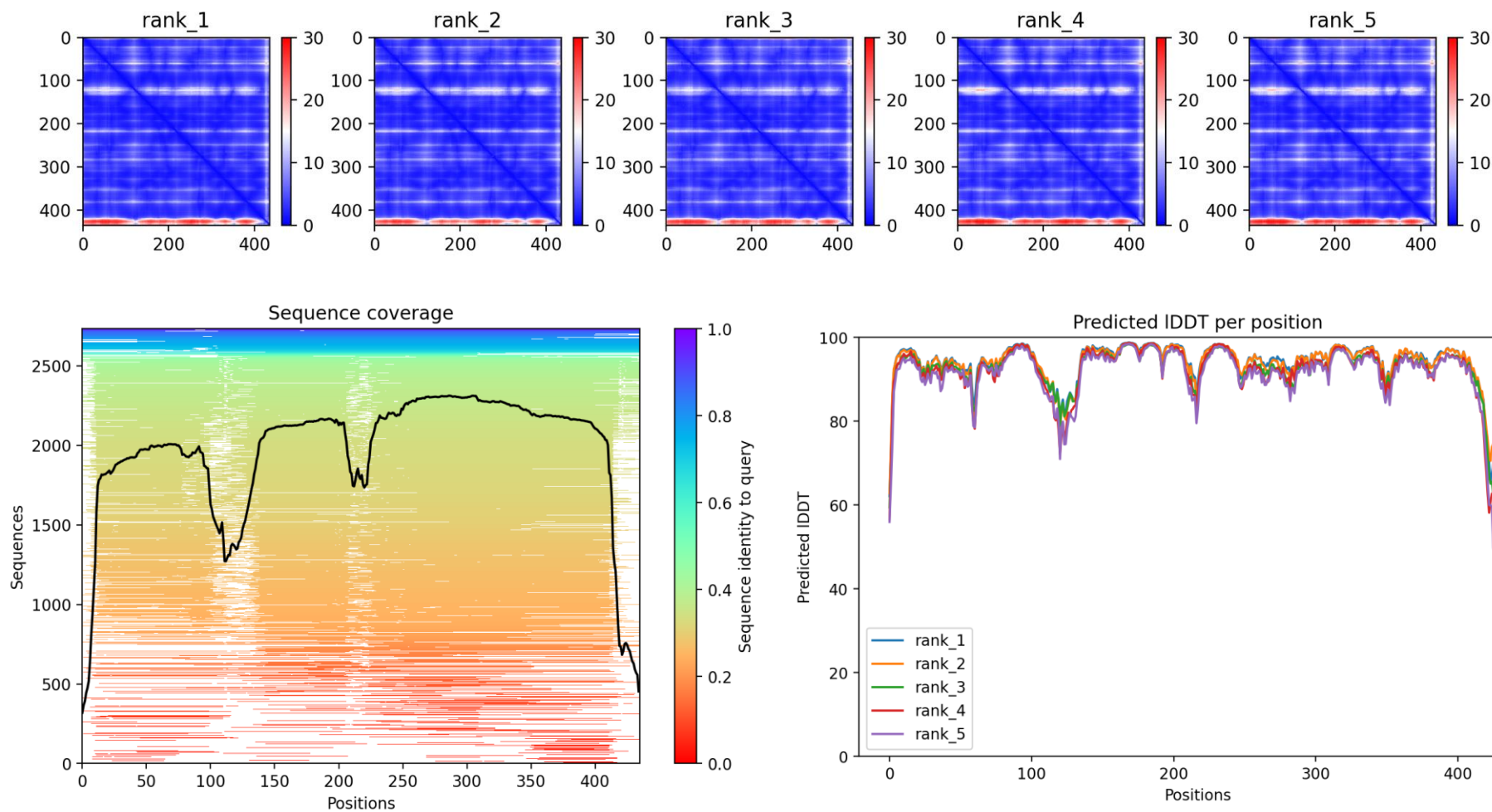
Superpose human myoglobin (PDB ID: **1MBO**) to one of the chains of hemoglobin (PDB ID: **4HHB**).

Exercise 2

Superpose the best model for PI3U protein to PI3U protein in the glycosylphosphatidylinositol (GPI) transamidase complex (PDB ID: **7W72**)



Exercise 1



PIGU (UniProt ID: Q9H490)
pLDDT 93.7



Outputs from ColabFold

- PDB files containing the predicted 3D coordinates of the protein or protein complex (pLDDT values are stored in the B-factor field) → can be analyzed with standard molecular visualization software packages (PyMOL, Chimera, Mol*)
- The MSA file (a3m format) -> can be visualized with any alignment viewer (check AlignmentViewer at MPI Bioinformatics Toolkit)
- MSA folder with found hits in different databases
- Sequence coverage plot
- Plots of the model quality (pLDDT plot and PAE plot)
- JSON file for each model which contains an array (list of lists) for PAE, a list with the per-residue pLDDT and the pTMscore.
- JSON file with predicted PAE scores
- Parameter log file (with avg. pLDDT).
- BibTeX file with citations for all used tools and databases.





Parameter configuration for ColabFold

ColabFold-AF2 has >15 tunable parameters (but don't worry, most can stay at their default values)

They might be useful for exploring AlphaFold2's full potential through:

- MSA-generation options
- MSA-sampling options
- Model prediction options

Overall, this can be useful for **improving structure prediction for challenging targets** or **sampling diverse structures**



Parameter configuration: MSA-generation options

msa_mode

- mmseqs2_uniref_env (default)
- mmseqs2_uniref
- custom (any type of alignment tool, Ex 2 of today's tutorial)
- single_sequence (for de novo-designed proteins)

▶ MSA options (custom MSA upload, single sequence, pairing mode)

msa_mode

mmseqs2_uniref_env

mmseqs2_uniref_env

mmseqs2_uniref

single_sequence

custom

Sh



Parameter configuration: MSA-sampling options

max_msa

controls the number of sequences used for structure prediction

deeper MSA will generally lead to a better prediction

lower values are useful when searching for alternative conformations

max_msa

auto

auto

512:1024

256:512

64:128

32:64

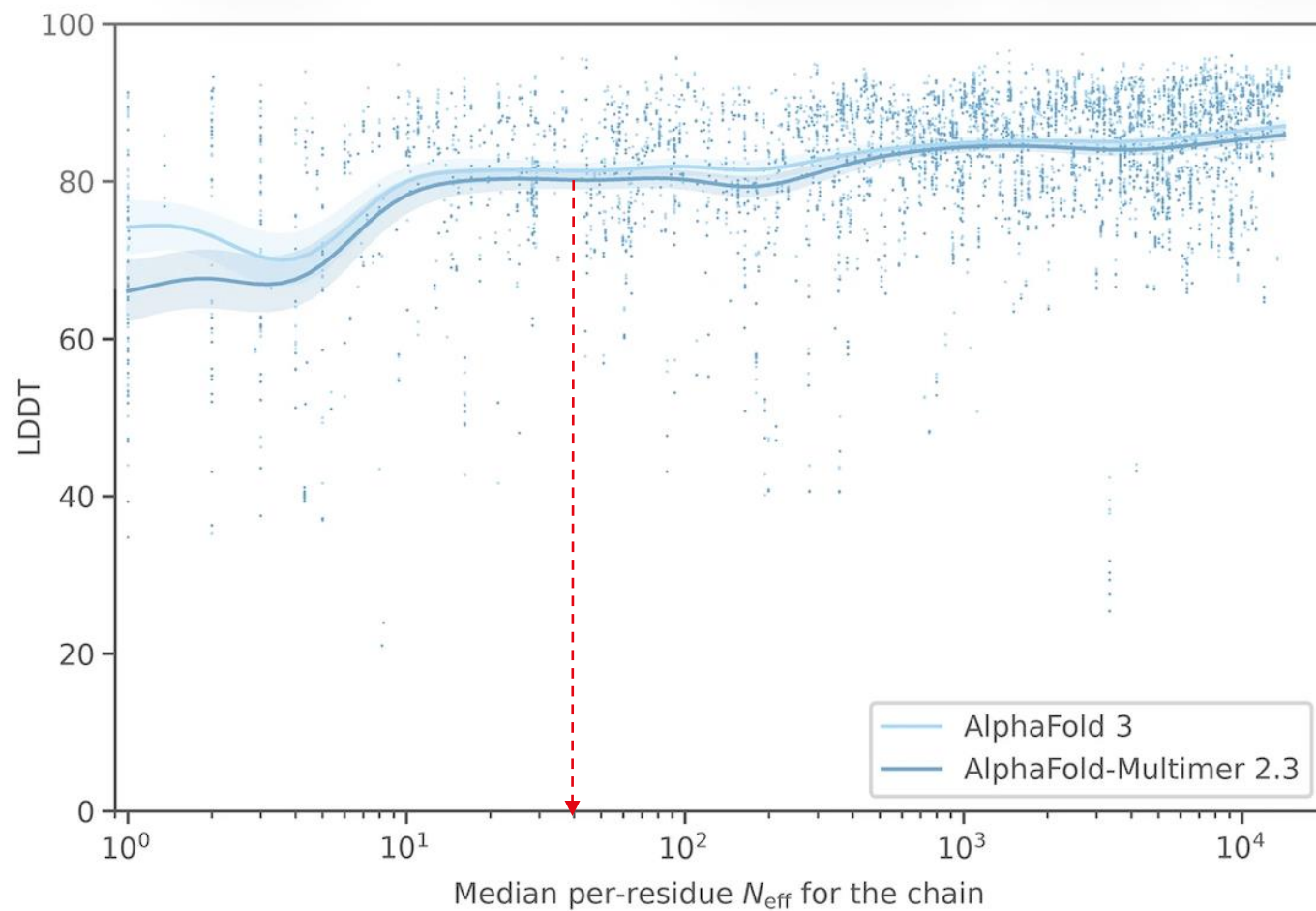
16:32

auto is 512:5120 (max_msa_clusters: max_extra_msa)



Parameter configuration: MSA-sampling options

What is a sufficient number of sequences in an MSA for a reliable prediction?



At least 30 diverse sequences (100 is better)



Parameter configuration: MSA-generation options

pair_mode: for heteromeric complex predictions

- unpaired_paired (pair and retain all sequences)
- Paired (only sequences that can be paired)
- unpaired

pair_mode

unpaired_paired

unpaired_paired

paired

unpaired

Sequence

chain A MAAPLVLVLVVAVTV:AATHLE
chain B VARGKRAALFFA

Paired MSA

F	Y	F	G	M	T	L	V	Y	C	T	A	Q	I	Y	L	V	I	D	L	Y	F	A	Y	I	K	R	E	F	C
F	Y	F	G	M	T	L	V	Y	C	T	A	Q	I	Y	L	V	I	D	L	Y	F	A	Y	I	K	R	E	F	C
F	Y	F	G	M	T	L	V	Y	C	T	A	Q	I	Y	L	V	I	D	L	Y	F	A	Y	I	K	R	E	F	C
F	Y	F	G	V	T	L	A	Y	A	T	A	Q	I	Y	L	V	I	D	L	L	F	A	Y	I	K	R	E	F	C

Unpaired MSA

F	Y	A	F	L	R	V	Y	L	T	H	G	L	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
F	Y	A	F	L	R	R	V	Y	H	L	T	H	G	L	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
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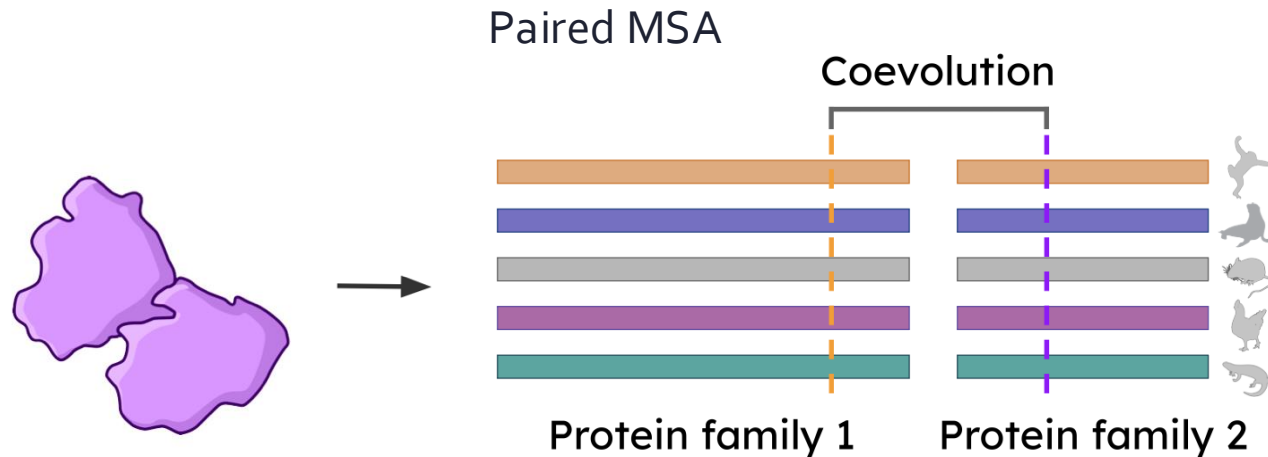


Parameter configuration: MSA-generation options

How is paired MSA built?

Hits are searched against UniRef100 and paired by the best hit with the same NCBI taxonomic identifier (species or subspecies)

Sequences are paired only when all query sequences are present for that taxonomic identifier



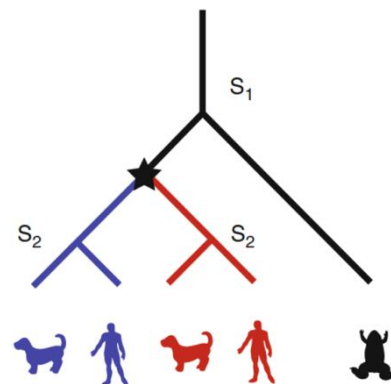
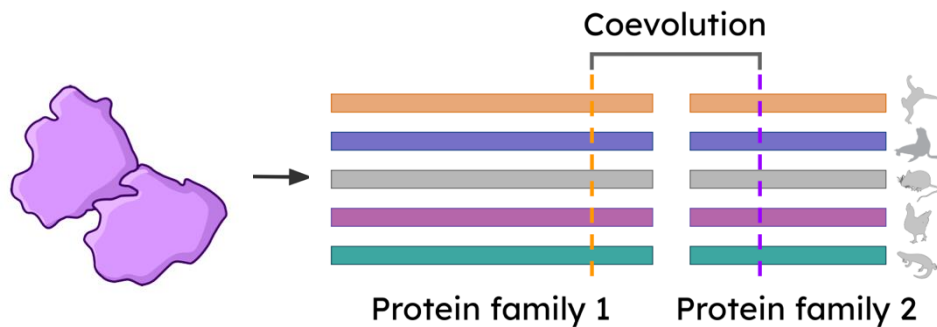


Parameter configuration: MSA-generation options

How is paired MSA built?

Hits are searched against UniRef100 and paired by the best hit with the same NCBI taxonomic identifier (species or subspecies)

Sequences are paired only when all query sequences are present for that taxonomic identifier



Paralogs genes separated by duplication
Orthologs genes separated by speciation

Yes BUT:

...only orthologous sequences carry true coevolution signal

...paralogs may add noise to the MSAs

...ColabFold's pairing strategy does not account for orthology/paralogy



Parameter configuration: MSA-generation options

pairing_strategy

- complete
- greedy

pairing_strategy

greedy

- **greedy** = pair any taxonomically matching subsets, **complete** = all sequences have to match in one line.

Example: if a complex has three chains and only two can be paired, it will be excluded from the paired MSA with complete strategy but included in the paired MSA with greedy strategy



Parameter configuration: MSA-sampling options

We can explore the stochasticity of AF2 through the following parameters

Sample settings

- enable dropouts and increase number of seeds to sample predictions from uncertainty of the model.
- decrease `max_msa` to increase uncertainty

max_msa

num_seeds

☐ use_dropout



Parameter configuration: MSA-sampling options

num_seeds:

- seeds determine random components throughout prediction (5 models per one seed)
- with max_msa, it can help explore alternative conformations
- setting a higher value can increase the chance of obtaining a better confidence score when the MSA is very small and templates are lacking

use_dropout:

activates dropout layers during inference, which prompts the AF2 neural network to be less confident in a single conformation

Sample settings

- enable dropouts and increase number of seeds to sample predictions from uncertainty of the model.
- decrease max_msa to increase uncertainty

max_msa

num_seeds

☐ use_dropout



Parameter configuration: model prediction options

model_type:

By default (auto) the best models, alphafold2_ptm (for monomers) and alphafold_multimer_v3 (for complexes), will be used

template_mode:

template_mode

none

none

pdb100

custom

Note:

If the MSA is deep (strong coevolution signal) AF2 tends to ignore template structures



Parameter configuration: model prediction options

num_recycles:

The number of times a prediction is re-fed to the model

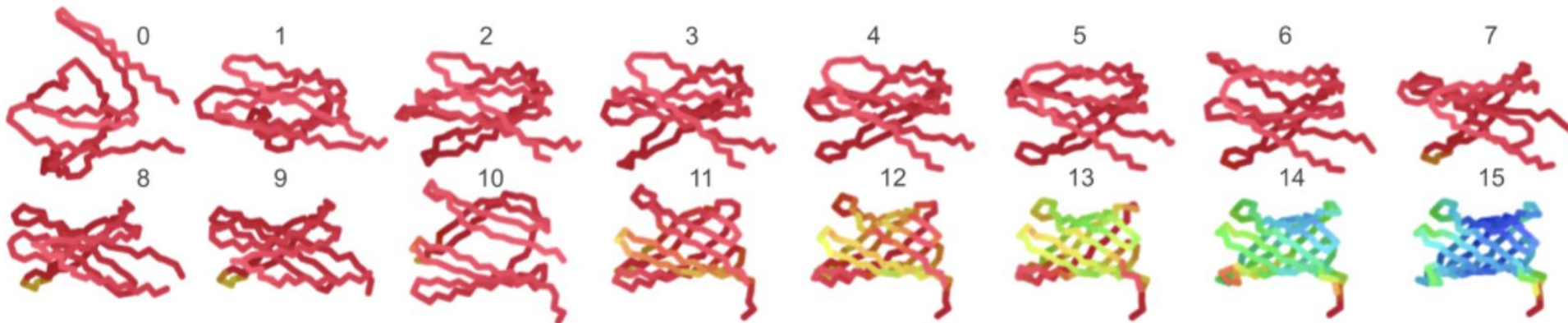
Higher values have been shown to improve predictions for some targets (normally from 3 to 20)

Decreasing the number of recycles to speed up the structure prediction

num_recycles

3

- if `auto` selected, will use `num_recycles=20` if `model_type=alphafold2_multimer_v3`, else `num_recycles=3`.





Parameter configuration: model prediction options

recycle_early_stop_tolerance:

avoid additional recycles or models if a sufficiently accurate structure was already found (save time, especially could be helpful for batch predictions)

recycle_early_stop_tolerance

auto

- if `auto` selected, will use `tol=0.5` if `model_type=alphafold2_multimer_v3` else `tol=0.0`.



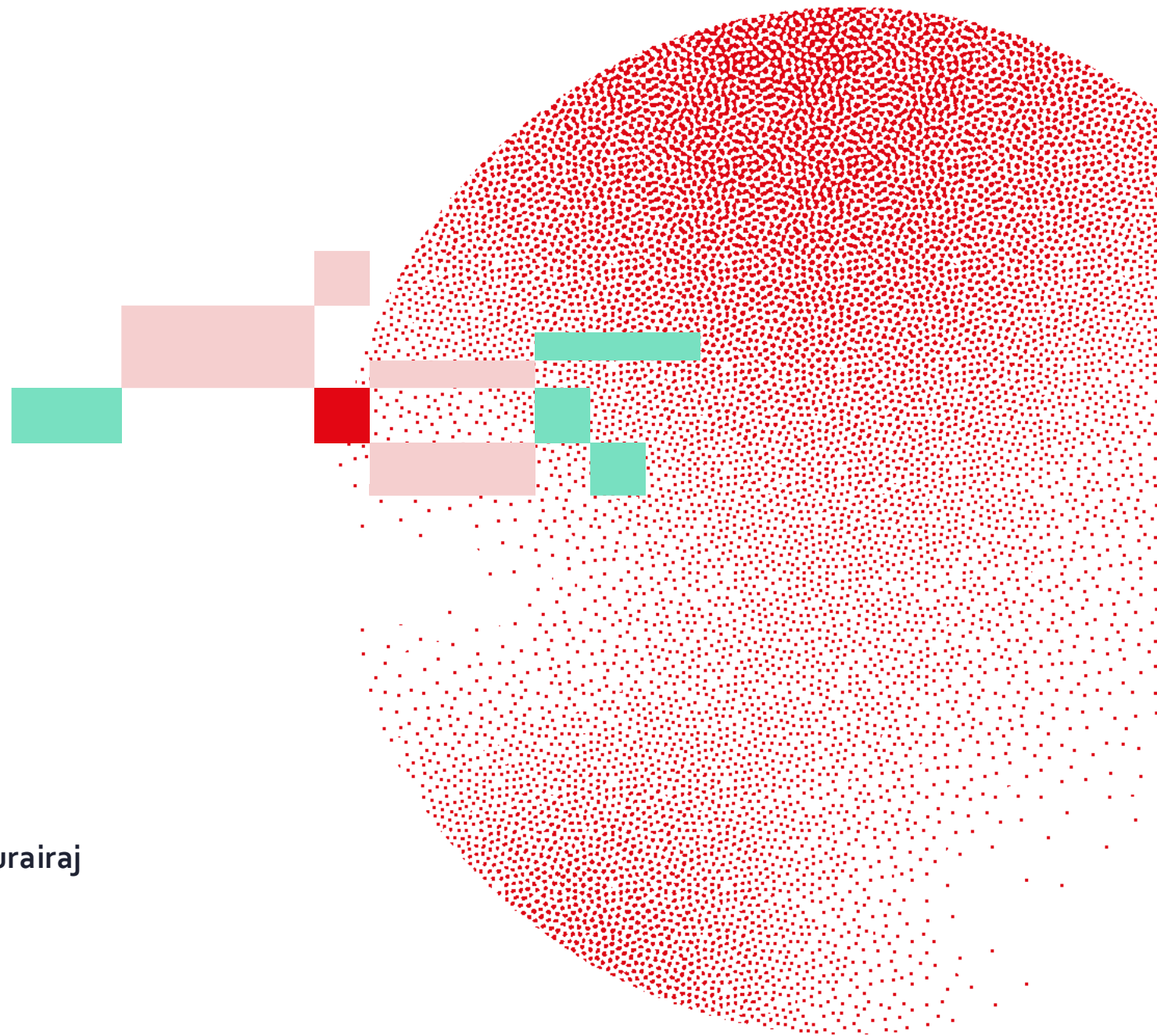
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DAY 1, PART 8

ColabFold tutorial Exercise 2

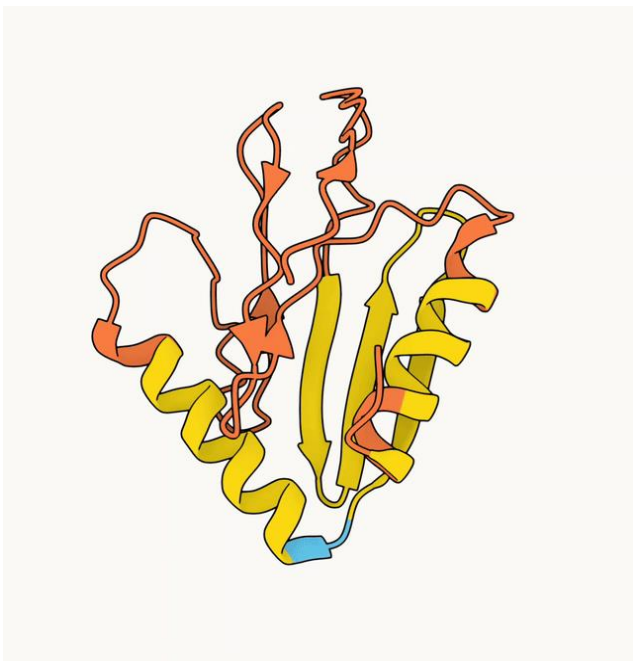
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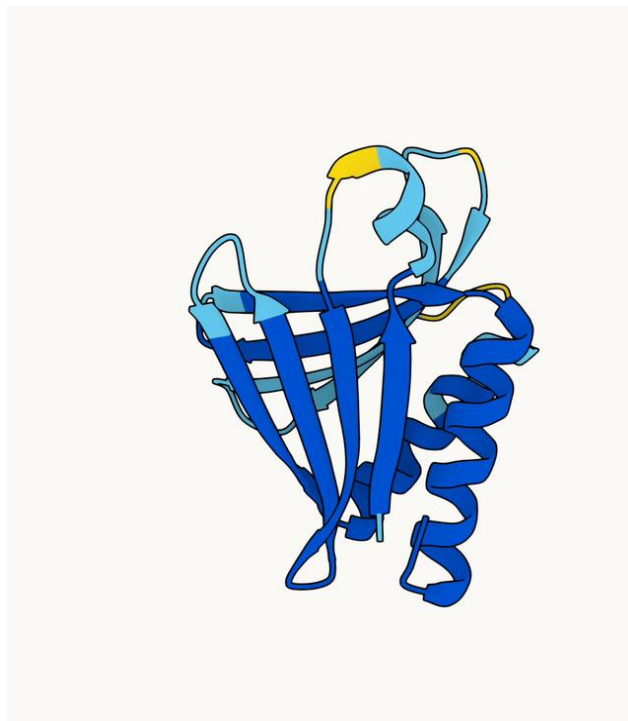




Exercise 2

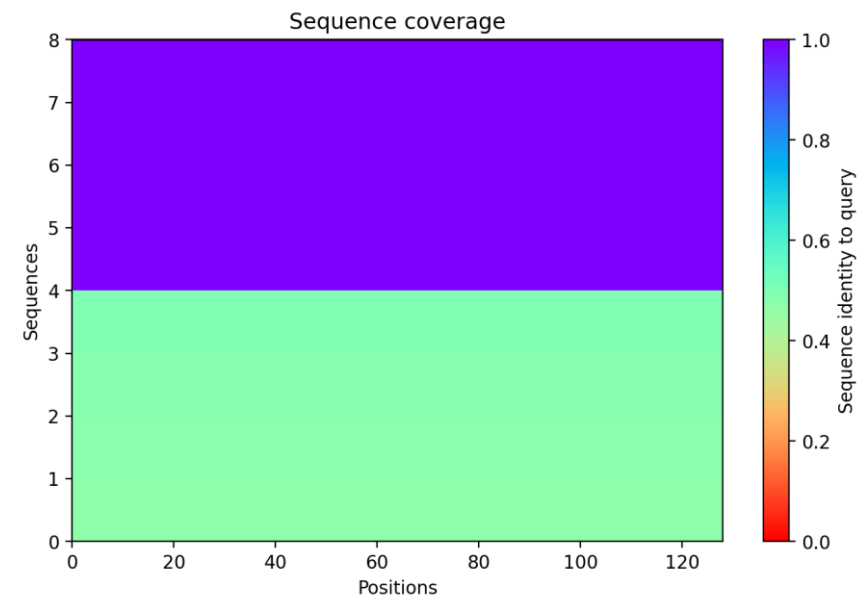


Default MSA
pLDDT **45.3**

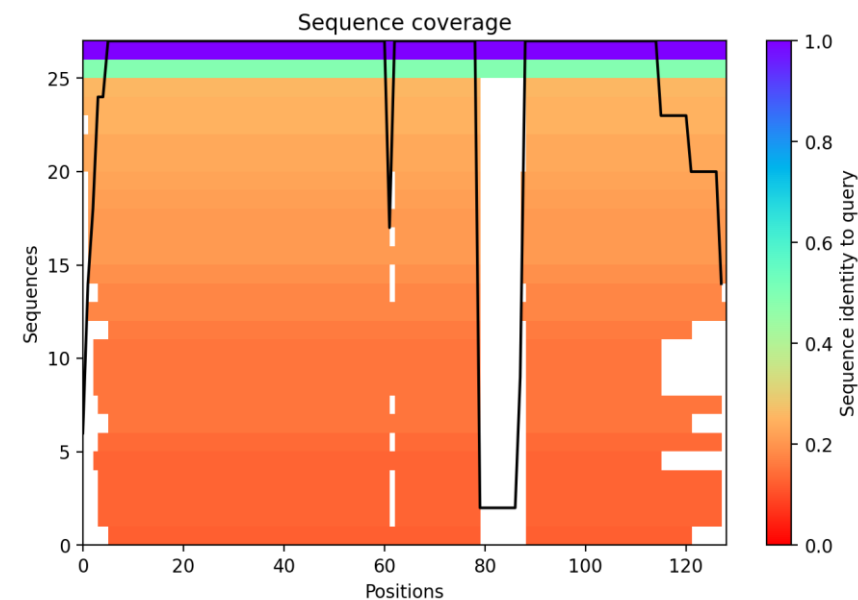


Custom MSA
PLDDT **89.2**

Default MSA



Custom MSA





Schedule

Day 1

Proteins, AlphaFold2 lecture

09:00 - 10:30

Coffee break

10:30 - 11:00

ColabFold lecture, tutorial 1

11:00 - 12:30

Lunch

12:30 - 13:30

ColabFold tutorial 2, Mol*

13:30 - 15:00

Coffee break

15:00 - 15:30

ColabFold tutorial 3

15:30 - 17:00



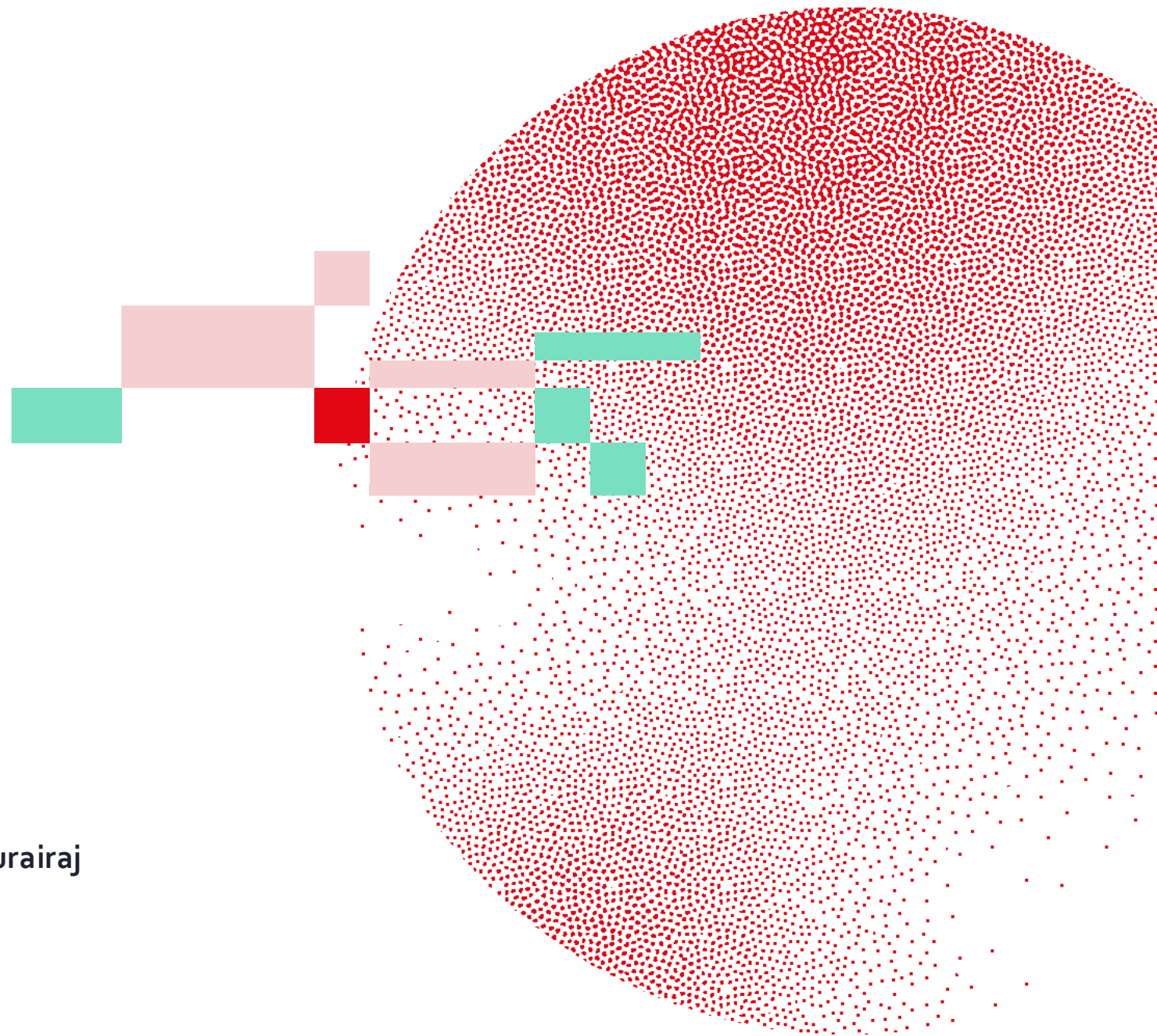
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DAY 1, PART 9

ColabFold tutorial Exercise 3.1

Diana Rapota, Rok Breznikar, Xavier Robin, Janani Durairaj

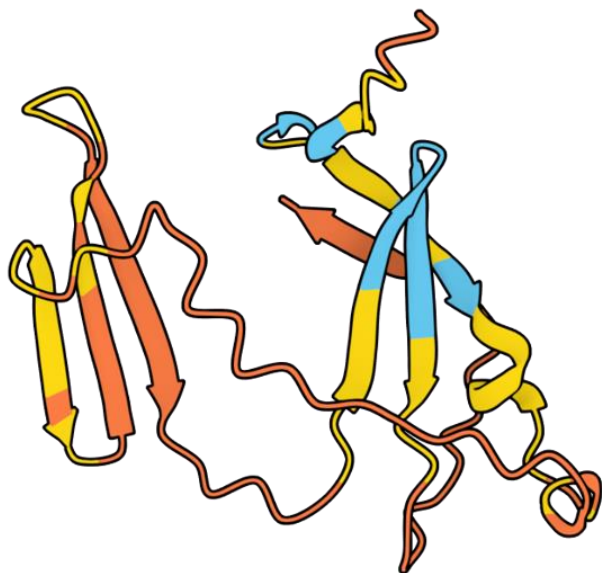
23-24 June 2026



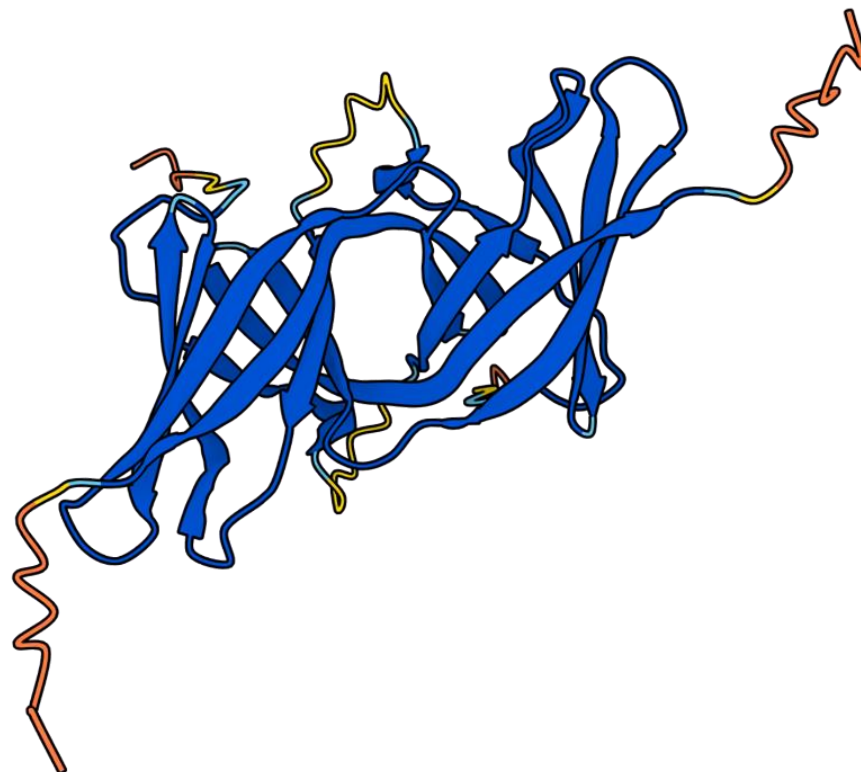


Exercise 3.1

A Q55DI5 (monomer)
pLDDT 50.6



B Q55DI5, homodimer
pLDDT=86 pTM=0.82 ipTM=0.82



AF2 models generated with ColabFold in which the target protein was modelled as a monomer(**A**) and as a homodimer (**B**)



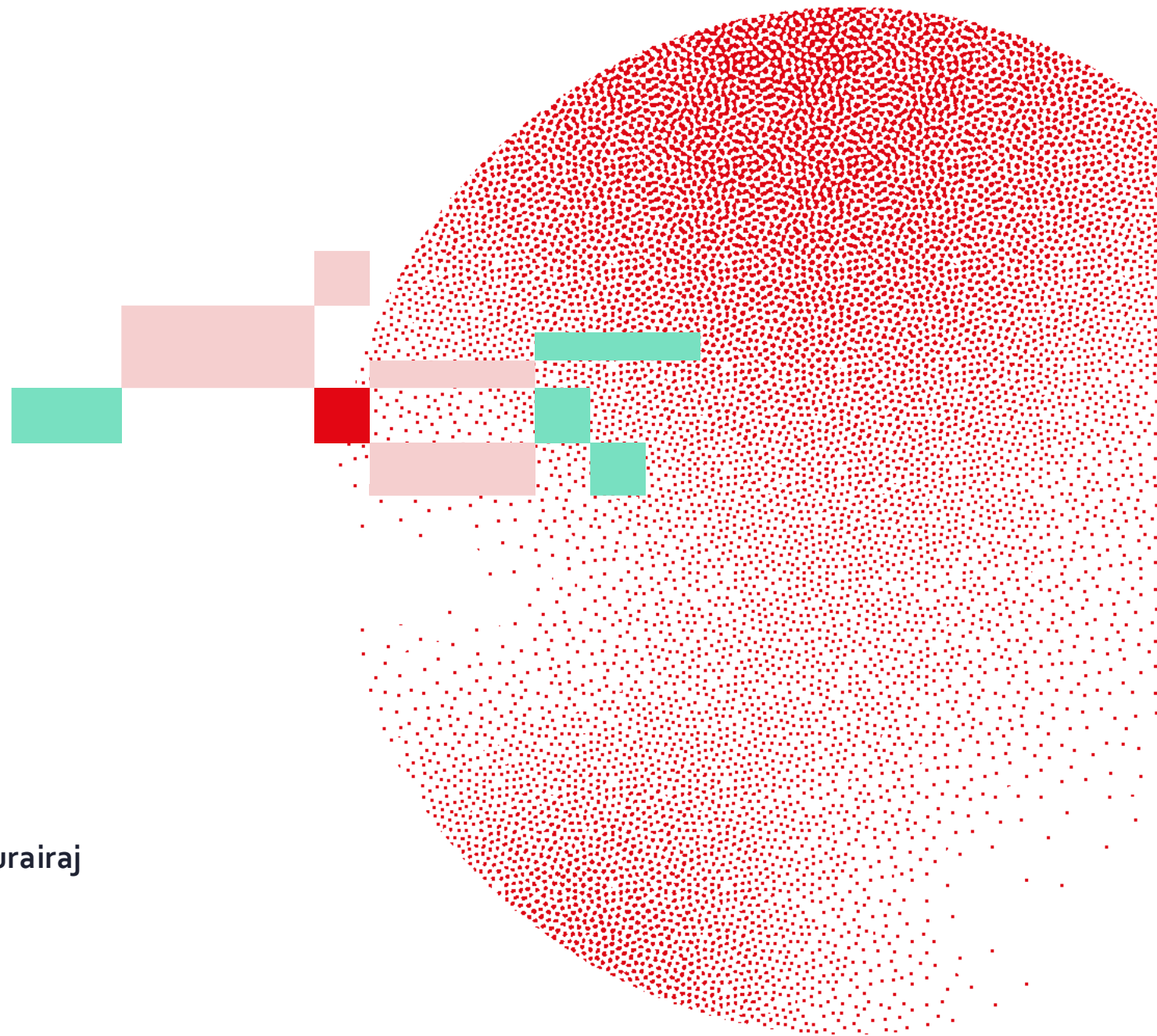
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DAY 1, PART 9

ColabFold tutorial Exercise 3.2

Diana Rapota, Rok Breznikar, Xavier Robin, Janani Durairaj

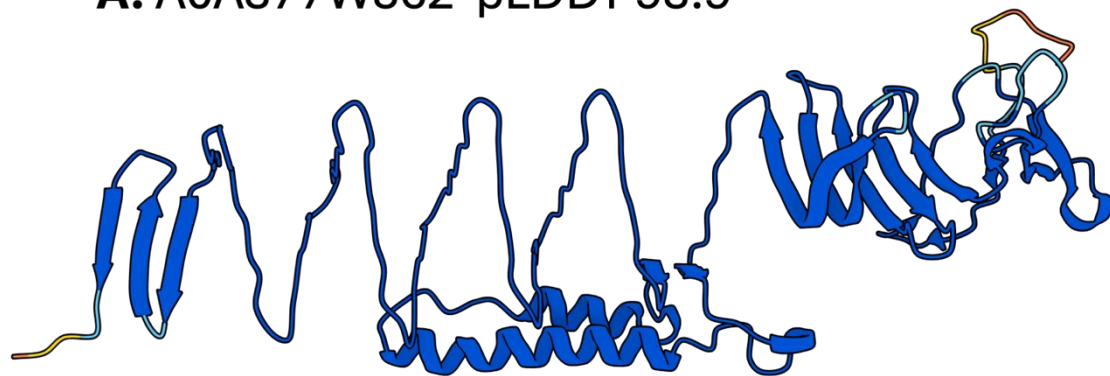
23-24 June 2026



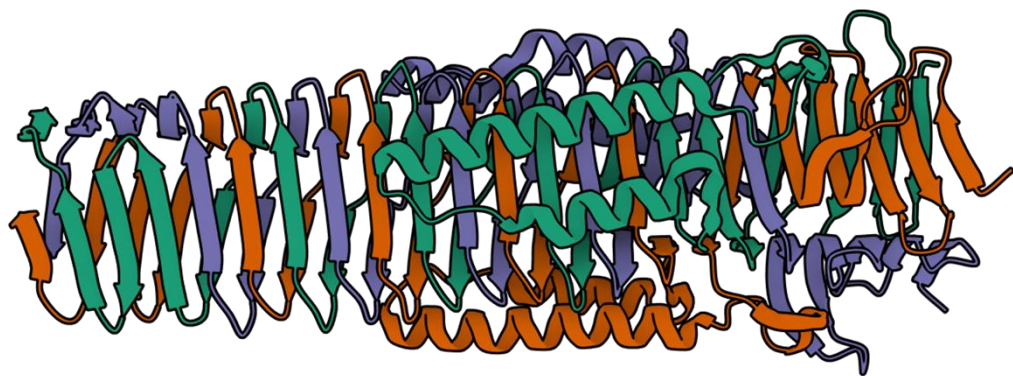


Exercise 3.2

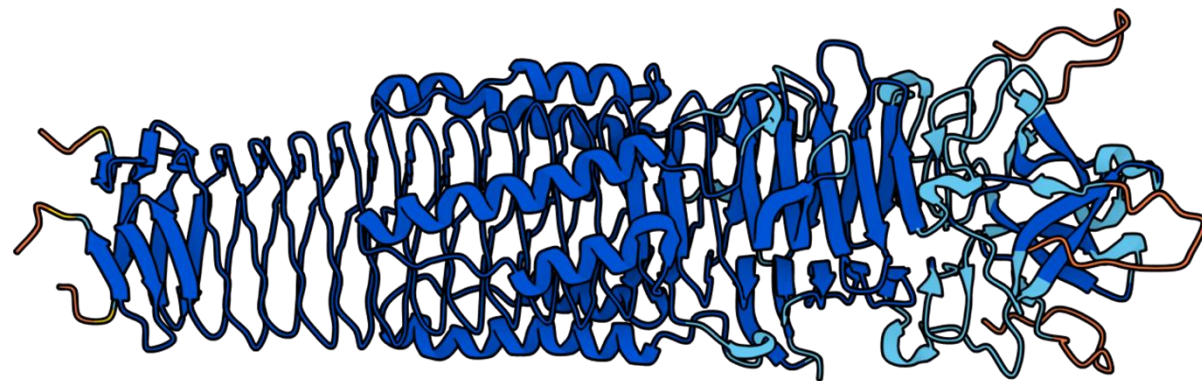
A. A0A377W562 pLDDT 93.9



B. PDB ID: 6SK0



C. A0A377W562 trimer pTM 0.93



A. AF2 model of the target protein as a monomer; **B.** PDB structure of a homologous obligate trimeric protein from the E. coli Type VI secretion system; **C.** AF2 model of the target protein as a homotrimer



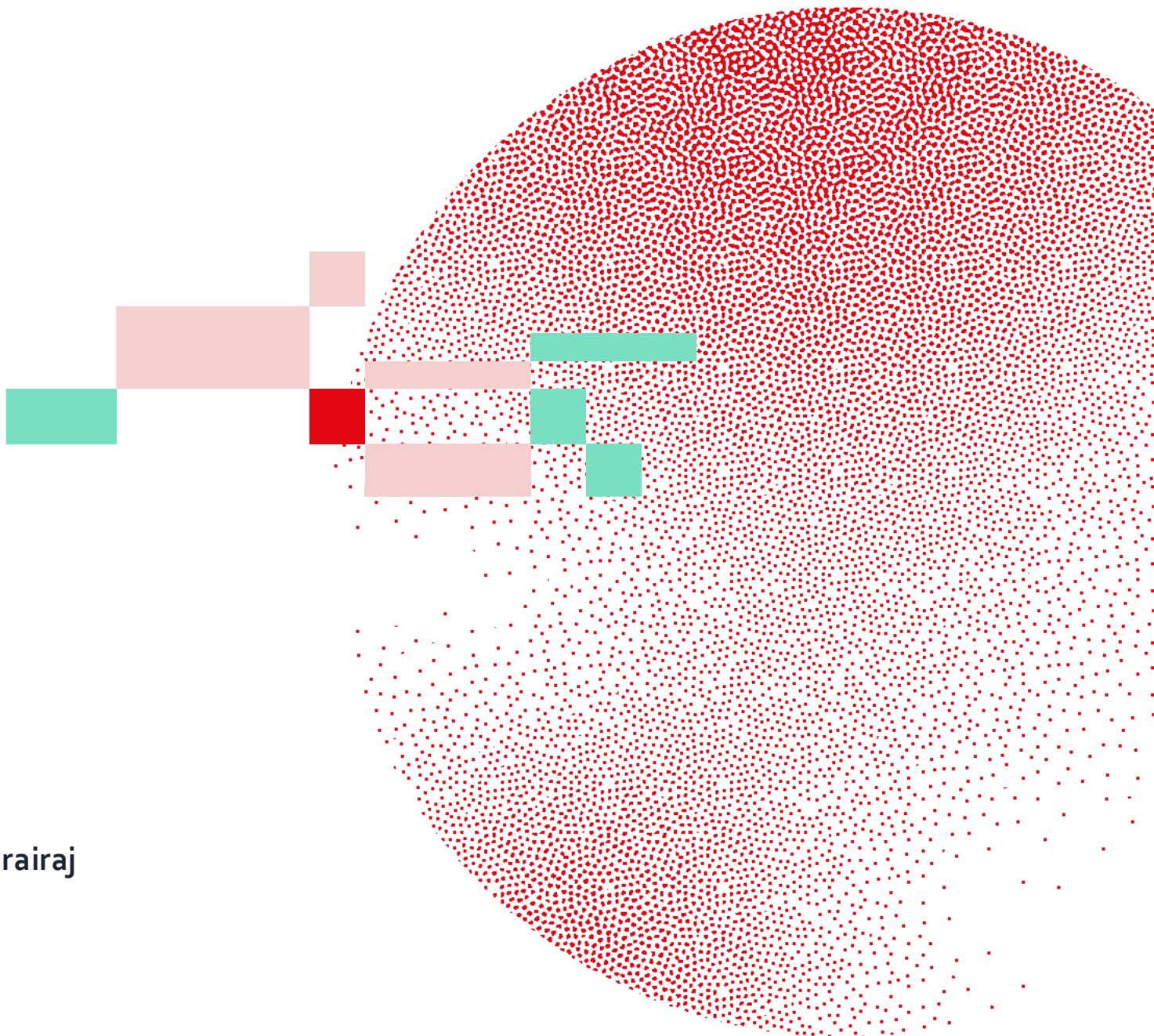
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DAY 1, PART 10

ColabFold tutorial Exercise 4

Diana Rapota, Rok Breznikar, Xavier Robin, Janani Durairaj

23-24 June 2026





Confidence when using templates but no MSA (single sequence mode)

- Models 1 and 2 use templates
- Models 3, 4 and 5 don't use templates

